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Optional Introduction to Jupyter and R

Introduction to Generalized Linear Models

Binomial Regression

Binomial Regression and Goodness of Fit

Ethical Issues in Statistics and Data Science

Reading: FairML Book, Introduction10 min

Peer-graded Assignment: Ethical Issues in Statistics and Data Science (Fair ML Intro)Grading in progress

Review Your Peers: Ethical Issues in Statistics and Data Science (Fair ML Intro)

Assignments: GLMs for Binomial Data

Peer-graded Assignment: Ethical Issues in Statistics and Data Science (Fair ML Intro)

🔔 It looks like this is your first peer-graded assignment. [Learn more](#)

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Review fellow learners

Congrats on submitting your assignment! Your peers can now review it and give you constructive feedback. You can do your part and help other learners complete this course by giving reviews.

Review assignments

Instructions

My submission

Discussions

Ethical Issues in Statistics and Data Science

Submitted on August 26, 2024

PROMPT

How are the models that we've used in this course so far (and in this specialization more broadly) based on the process of induction?

The models used in this course and specialization are based on the process of induction because they emphasize on **generalization** of the machine learning model (trained on the observed training dataset), measured by testing the prediction performance (e.g., accuracy) on the **unseen** (test) dataset.

The algorithms are trained on a dataset to recognize patterns, relationships, and correlations, which are then generalized to make predictions or decisions on new, unseen data. Also, special care (e.g., **regularization**) is taken to prevent **overfitting** of the model on the training dataset (addressing to the classical **bias-variance tradeoff**), so that the models don't capture noisy patterns from the training dataset.

This inductive approach assumes that the patterns in the training data will hold true in future scenarios. The process involves using specific examples (data) to infer general rules or models, which is the essence of induction.

PROMPT

"The fact that machine learning is 'evidence-based' by no means ensures that it will lead to accurate, reliable, or fair decisions." Provide 2-3 examples of this claim.

Machine learning models are only as good as the data they are trained on. If the training data is biased, the model will likely reproduce and even amplify these biases, that will fail to ensure accurate, reliable or fair predictions. Let's dive into the following examples in support of the claim:

1. Bias in Training Data: For instance, if a **hiring algorithm** is trained on historical hiring data from a company that has traditionally favored a particular demographic, the algorithm may continue to favor that demographic, leading to unfair and biased hiring decisions.

2. Unfair Predictive Assumptions: Consider a **predictive policing model** that uses historical crime data to forecast future crime hotspots. If the historical data is skewed due to over-policing in certain neighborhoods, the model might inaccurately predict high crime rates in those areas, leading to a self-reinforcing cycle of over-policing and marginalization of certain communities.

3. Representation Harm: An example is **image search algorithms** that reinforce gender stereotypes by predominantly showing men as CEOs or software developers. Although these algorithms are based on evidence (existing images and patterns), they perpetuate harmful stereotypes that can influence societal perceptions and contribute to discrimination. Also, a **machine translation system** reflects the social gender bias and prejudice by using the pronoun 'He' for doctors and 'She' for nurses, when translating from a gender-neutral language back to english.

PROMPT

Do you believe that Amazon's same-day delivery system mention on page 3-4 is unfair or unjust? Why or why not?

As described in the book, Amazon's same-day delivery system can have potential impact on different communities. Whether it is unfair or unjust depends on how the service is distributed across different demographics.

If the service disproportionately excludes certain neighborhoods, particularly those with higher populations of marginalized groups, it could be considered **unjust** or **unfair**. This exclusion could reinforce existing inequalities by providing more convenience and economic opportunities to already privileged areas while neglecting underserved communities.

If the algorithm determining delivery areas is based on criteria that indirectly discriminate against certain groups, it would be an example of an **unfair** application of machine learning.

PROMPT

What is the machine learning loop? Do you think that the machine learning loop, given in figure 1, applies to statistical modeling? Justify your answer.

The machine learning loop refers to the cyclical process in which machine learning systems operate. This loop involves the stages of measurement, learning, action, and feedback. In this process, data is collected (measurement), a model is trained on the data (learning), the model makes predictions or decisions (action), and the outcomes of these decisions affect the environment, which generates new data that is fed back into the system (feedback).

Yes, the machine learning loop can apply to statistical modeling as well, although there are differences. In statistical modeling, data is also collected and used to fit a model, which then makes predictions or inferences. The outcomes can influence future data collection or model refinement, creating a feedback loop similar to that in machine learning. However, the key difference lies in the emphasis on automation and **continuous** learning in machine learning, whereas statistical modeling often involves more **static** analysis with explicit human intervention at each stage. Nonetheless, the concept of iterative refinement and the influence of outcomes on future predictions is common to both fields.

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SD

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